Toward Mechanical Scientific Models: Fractal Custody and Governed Agent Experiments

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Abstract

The *Mechanical Scientific Method* treats scientific computation as a sequence of governed state transitions in which evidence, transforms, and bounded claims remain inspectable rather than collapsed into model prose. We propose the term *Mechanical Scientific Model* for a computational model class that satisfies explicit requirements: explicit evidence state; governed and versioned state transitions; separation of deterministic transforms from probabilistic model outputs; preservation of positive, null, negative, failed, blocked, timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse evidence traceability; and successor-state recording rather than silent history rewriting. *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG) provide the custody substrate for binding those transitions. HydraDG is the primary experimentally evaluated implementation examined here: it preregisters hypotheses, freezes case manifests and scorers, records raw prompt and response hashes, and preserves terminal states without silent substitution. We report two preregistered HydraDG studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval for failure-prevention endpoints; both terminate as *underpowered*, so confirmatory treatment-effect claims are not supported while null, negative, and failure evidence remains in custody. We do not claim that Mechanical Scientific Models are an established external field term; we offer them as a proposed model class and report bounded custody behavior from one implementation.

1 Introduction

Large language model agents are evaluated on task success, benchmark scores, and human preference [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models, stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative and null cells disappear from the public record, later work cannot distinguish *no effect* from *no measurement*.

This paper separates five layers that are often conflated in agent-evaluation reporting:

- **Method** — the *Mechanical Scientific Method*: governed transitions among explicit evidence states rather than narrative collapse of runs into conclusions.
- **Model class** — *Mechanical Scientific Models* (proposed here; not an established external terminology): computational systems whose state transitions are mechanically exposable and failure-complete.
- **Custody substrate** — FCO/FCG receipts that bind source bytes, transforms, derived evidence, bounded claims, and successor state.

- **Primary implementation** — HydraDG, the experimentally evaluated implementation whose preregistered EXP-008 and EXP-009 results are reported here.
- **Systems validation and companion exemplars** — HydraLamp perturbation receipts and HydraDG scoring on local Apple Silicon (Mac Studio class, 32 GB RAM); Antigence anticube lanes (supplement F11–F12); Vithia companion lineage in repositories `vitaology` and `fractal-custody-objects` (`ZERO_PRIMARY_WEIGHT` per cross-repo census; companion runs not vendored into this submission checkout).

HydraDG instantiates the proposed model class through a custody-first experimental contract. Every actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic model outputs are never treated as authority; they are evidence objects subject to deterministic scoring and graph append rules.

This paper reports terminal results from the IC Failure Learning structured-retrieval falsification lane executed in HydraDG. Primary numbers are reverse-traced to frozen preregistered verdict JSON receipts; companion tables and figures cite SHA-256–bound supplement artifacts bundled with this submission. We do *not* claim that structured FCG context improves failure prevention: EXP-008 and EXP-009 closed as *underpowered* (`n_paired`= 2, `discordant`= 0) with `conclusion_bounded=effect not established`. Exploratory secondary patterns and vitaology runs are not promoted to primary findings for those experiments.

2 Related Work

Agent evaluation. Benchmark suites measure capability but rarely require publication of failed runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks with mandatory negative capture at the custody layer.

Retrieval and structured context. Retrieval-augmented generation and graph-augmented memory systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null retention, and power-aware verdicts; HydraDG enforces both.

Local open-weights evaluation. Frozen EXP-008/009 cells use `qwen3:1.7b` and `qwen2.5-coder:7b` only; LongMemEval track-03 rows in supplement table A4 use `qwen2.5-7b` and `deepseek-r1-14b` from `eval/track_model_k_20260820/` with claim ceiling `HISTORICAL_SUPPORTING_ZERO_PRIMARY_WEIGHT`. No frontier API model executed a preregistered primary scorer in this submission.

Provenance, integrity, and reproducibility. W3C PROV-O and workflow provenance systems represent process and research-object lineage [12, 13], while append-only Merkle audit logs provide established tamper-evident integrity mechanisms [11]; HydraDG does not claim to invent these primitives. Recent work surveys evidence tracing and execution provenance for LLM agents [15], and separate work argues for preserving negative ML results rather than selective omission [14]. Preregistration, FAIR principles, and nanopublication-style lineage further support reproducibility [5–7]. HydraDG integrates rather than replaces these foundations: probabilistic outputs, deterministic transforms, malformed cells, failures, abstentions, blocked states, and underpowered terminations remain typed evidence with explicit claim-promotion ceilings enforced via FCO/FCG handoff receipts. Companion custody-framework preprints (deferred to camera-ready de-anonymization) define FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results reported here.

3 Framework

3.1 Mechanical Scientific Models

We propose the term *Mechanical Scientific Model* for a computational scientific system whose internal evidence-state transitions are mechanically exposable rather than inferred post hoc from model prose. We do not claim that this term is already established in the literature; it names a proposed model class whose required properties can be checked against an implementation.

Let S_t denote the governed evidence-and-claim state at step t. A Mechanical Scientific Model advances by governed transforms

$$S_t \xrightarrow[E_t]{T_t} S_{t+1},$$

where T_t is a versioned, auditable transform (deterministic scorer, ingest pass, governed model call, or tool invocation) and E_t is the evidence bound to that transition. Every transition must emit or bind:

1. source or upstream evidence;
2. transform, tool, or model identity (with deterministic vs. probabilistic output separation);
3. derived evidence or parser state;
4. a bounded claim with an explicit ceiling; and
5. an artifact or state delta recorded as successor state rather than silent history rewriting.

Required properties include: explicit evidence state; governed and versioned state transitions; deterministic vs. probabilistic output separation; preservation of positive, null, negative, failed, blocked, timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse evidence traceability; and successor-state recording. Figure 1 situates the Mechanical Scientific Method, FCO/FCG custody, the proposed model class, HydraDG as the evaluated implementation, Vithia as a companion exemplar, and execution components used in this submission.

3.2 Custody objects

An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call, scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges between FCOs with hash-linked roots [11]. FCO/FCG provide the custody substrate on which Mechanical Scientific Models record transitions; HydraDG is one implementation that preserves failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared at preregistration and enforced at promotion time.

3.3 Anticube evidence typing

Portfolio anticube semantics classify evidence along established vs. exploratory and positive vs. negative axes rather than a single success bit. In this checkout, Antigence-derived anticube state appears in supplement figures F11–F12 and table `A7_ANTICUBE_SOT_STATE_LEDGER`; temporal trajectory coordinates are sparse and labeled `NOT_IN_CHECKOUT` where source graphs are absent. These panels document classification mechanics only; they do not upgrade EXP-008/009 claim ceilings.

3.4 Experimental freeze

Each experiment declares:

- frozen case manifest and prompt-contract hashes;
- conditions (e.g., flat prose vs. structured FCG);
- models with expected runtime digests;
- primary endpoint and aggregation rule;
- explicit gates for confirmatory vs. exploratory claims.

Execution emits `START_RECEIPT`, per-cell raw outputs with prompt/response SHA-256, parser state, and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substitution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as terminal evidence.

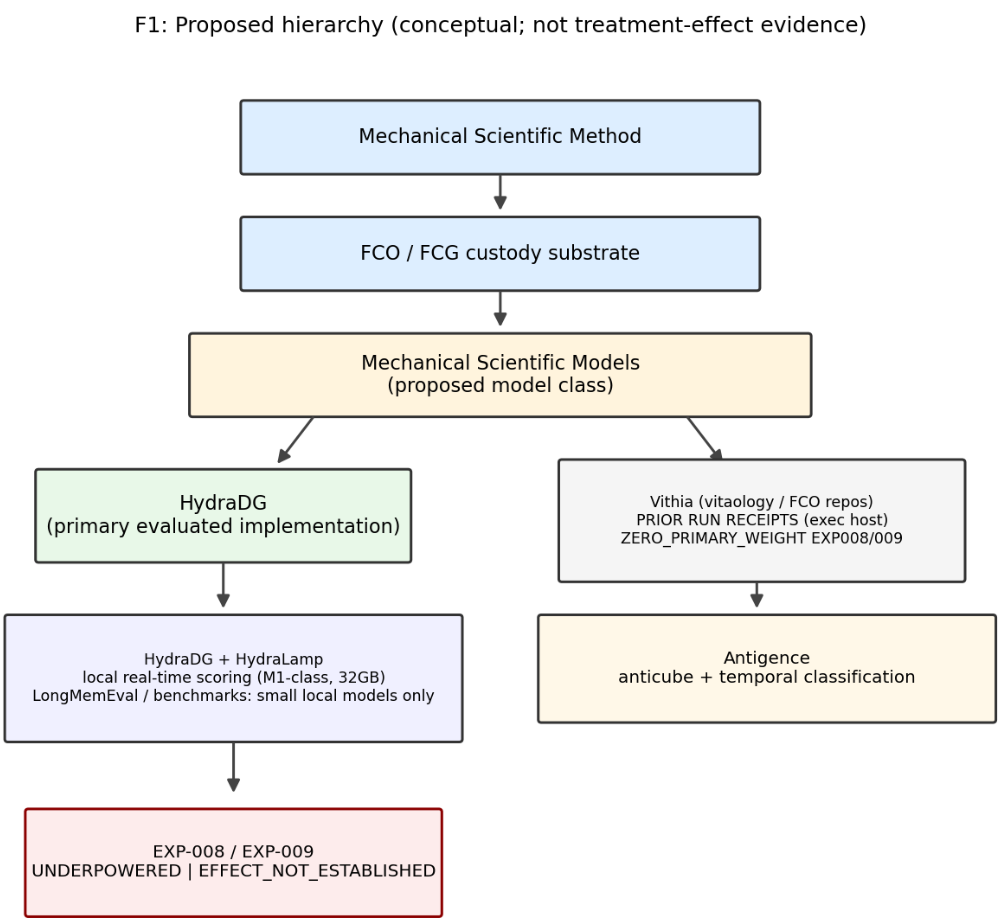

Figure 1: Proposed intellectual hierarchy (conceptual; not empirical effect evidence). HydraDG is the primary experimentally evaluated implementation; Vithia maps to companion repository `vitaology` (runs retained on the scientific execution host, `NOT_IN_CHECKOUT` here) and `fractal-custody-objects`; Antigence anticube panels are supplement-only; EXP-008/009 remain *underpowered*.

3.5 Governed local execution

Where preregistered, local model calls route through a governed bridge that records selection age, swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate evidence classes; one is not silently labeled as the other. EXP-008/009 executed on an Apple Silicon Mac Studio class workstation (32 GB RAM) with HydraDG scoring and HydraLamp systems-validation receipts on the same host class.

3.6 Evaluation model policy

All preregistered scientific endpoints in this submission use small local open-weights models only (`qwen3:1.7b`, `qwen2.5-coder:7b`; LongMemEval historical lanes per A4 matrix). Frontier coding agents assisted manuscript preparation only; they did not execute frozen primary scorers or replace verdict-bound cells.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

4 Experimental Setup

4.1 Task family

The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen `CASES.jsonl` manifest; each case receives multiple replicates per condition.

4.2 Conditions and models

C0 (FLAT_PROSE): failure-handling guidance as unstructured text.
C1 (STRUCTURED_FCG): the same semantic content composed through a frozen FCG retrieval contract.

EXP-008 and EXP-009 used local models `qwen3:1.7b` and `qwen2.5-coder:7b` under identical case and scorer contracts on private M1-class hardware (32 GB RAM). Thinking/reasoning modes were held consistent with preregistration. Primary aggregation: majority of three replicates per case. Historical LongMemEval track-03 retrieval ablations (supplement F4–F5) replay the same local-only policy with `qwen2.5-7b` and `deepseek-r1-14b`; they are labeled `HISTORICAL_SUPPORTING_ZERO_PRIMARY_WEIGHT` and do not promote primary claims here.

4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Figure 2 decomposes the same receipts into attrition counts and `data_quality` rates ($n_{\mathrm{raw}} = 300$ per study). Both experiments executed the full raw matrix with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

EXP-009. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

Stage-2 baseline. Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established `FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED`, providing historical context for the structured-retrieval falsification lane. LongMemEval track-03 historical lanes (local `qwen2.5-7b` / `deepseek-r1-14b`, frozen primary scorer) are retained in the supplement for

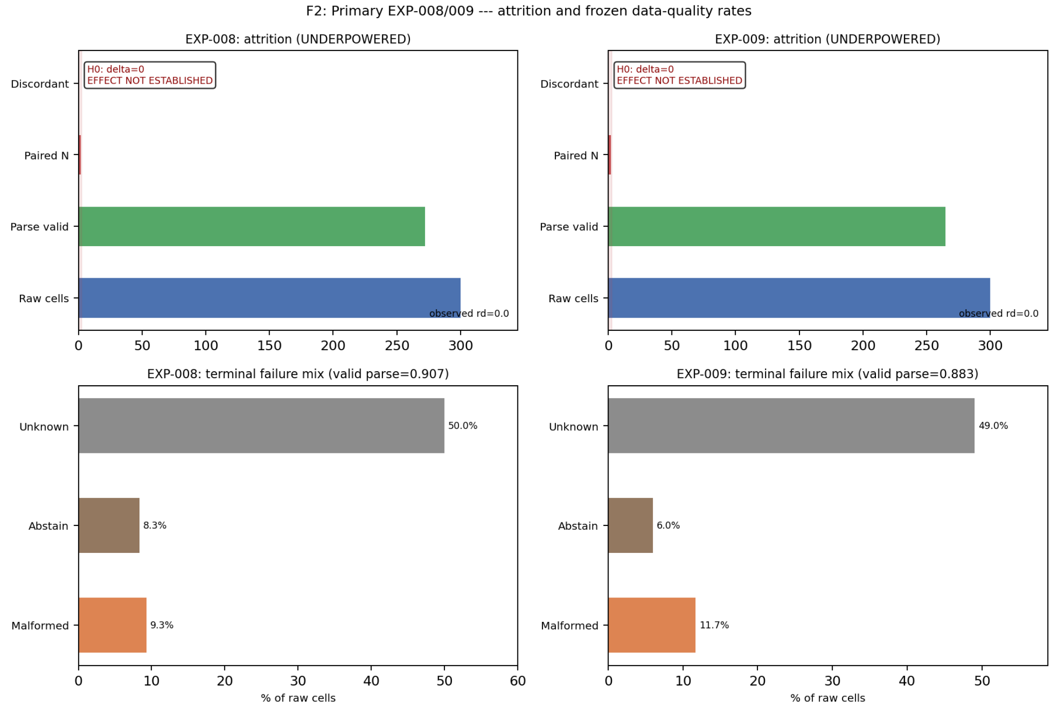

Figure 2: Primary EXP-008/009 outcomes from frozen verdict JSON: top panels show attrition (raw → paired); bottom panels show terminal failure rates (malformed, abstain, unknown). H0 reference shown; UNDERPOWERED — effect not established (not proof of null).

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

audit only; they were not executed with frontier models and do not upgrade primary EXP-008/009 claims.

5.1 Failure-preserving systems validation

The preregistered retrieval experiments above test whether structured context improves a frozen endpoint. A separate systems-validation suite asks a different question: does the custody mechanism preserve integrity under perturbation, tampering, concurrency, replay, and external execution failure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix; these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context improves the EXP-008/009 endpoint.

The tamper suite used synthetic cases only (not real security incidents). In a live provider repair ladder, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure remained a terminal auditable state rather than being erased or substituted. The same custody primitives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-document atomization, offline anomaly detection (Antigence anticube lanes), bounded bioinformatics, substrate generation, and operator-facing scientific workflow monitoring; these im-

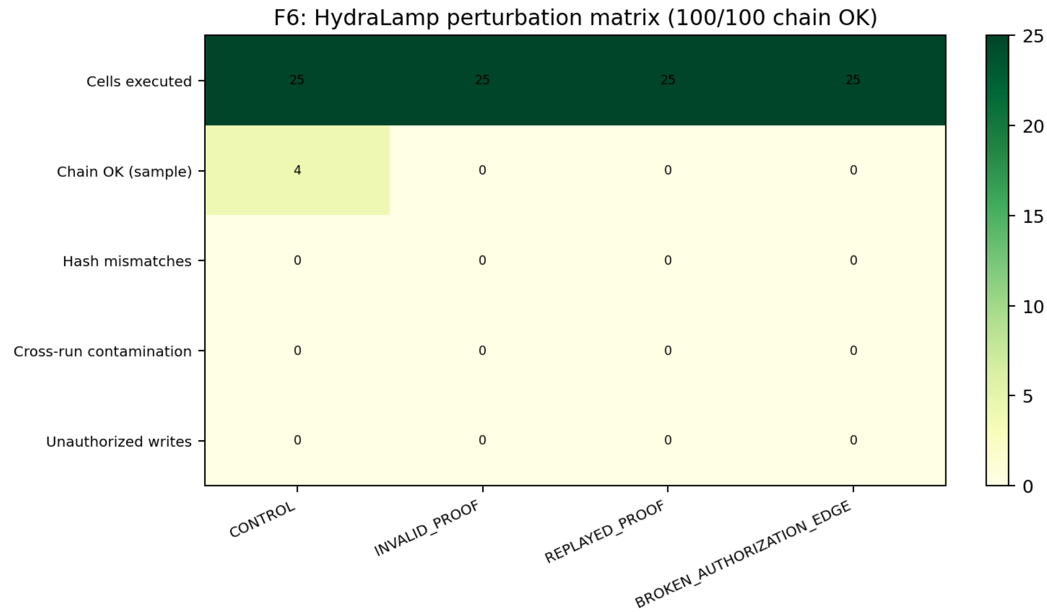

Figure 3: HydraLamp perturbation test matrix (systems robustness only; not treatment-effect evidence).

plementations motivate transfer studies but do not provide additional evidence for the treatment effect reported here.

6 Discussion

The principal contribution of this submission is not evidence that structured FCG retrieval improves failure prevention. EXP-008 and EXP-009 remain *underpowered*; the empirical retrieval effect on the preregistered E06 endpoint is *not established*. The broader methodological proposition is that scientific computational models can expose their evidence-state transitions mechanically—through governed receipts, explicit claim ceilings, and preserved terminal states—so that null, failure, and blocked outcomes remain part of model state rather than disappearing from the public record.

HydraDG demonstrates bounded custody and systems-validation behavior under that contract: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. HydraLamp perturbation, tamper, concurrency, and replay results (Table 2) evaluate custody mechanics only. Cross-repo census row `vitaology | EXP-001..014` is admitted as `RELATED_IMPLEMENTATION` at `ZERO_PRIMARY_WEIGHT`; vitaology run receipts are retained on the scientific execution host and are not vendored into this submission checkout (`NOT_IN_CHECKOUT`). Antigence anticube panels (F11–F12) type evidence for audit; they are not confirmatory treatment-effect evidence for EXP-008/009. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

6.1 Limitations

Limitations include bounded case replication, deliberate local-only small-model lanes in the reported matrix (no frontier API substitution), and absence of runtime-equivalent successor replication in primary results. Execution hardware was constrained to M1-class Apple Silicon with 32 GB RAM; results do not establish performance on datacenter GPU fleets or frontier-model endpoints. A preregistered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final Parquet serialization without a `BUILD_RECEIPT`; partial node/edge artifacts are not readback-safe, so terminal whole-corpus atomization is not established in this submission.

Biological, protein-structure, and cellular perturbation applications of the custody framework remain future validation targets, not demonstrated generalizations of the agent-context results here.

6.2 Future directions

Agent systems. Larger successor replications, governed multi-agent handoffs, selective replay, and resource-aware custody gates under preregistered manifests.

Structured retrieval at scale. Checkpointed ingest and transactionally durable atomization so long hierarchy builds can resume from batch boundaries rather than materializing the entire graph at finalization.

Cross-project federation. Project-local FCG roots with explicit cross-project references and no automatic evidence admission between repositories.

Cryptographic custody. Authorized signatures and MMR commitments where policy permits; hashes remain identity commitments unless a verified signing operation occurred.

All directions above are *planned* unless independently measured; they do not extend the EXP-008/009 empirical claims.

7 Conclusion

We proposed the term *Mechanical Scientific Model* for a computational model class whose governed state transitions expose evidence, separate deterministic from probabilistic outputs, preserve failure-complete terminal states, and bind explicit claim ceilings with forward and reverse traceability. FCO/FCG provide the custody substrate; HydraDG is one implementation examined here under that proposal. Two preregistered HydraDG studies (EXP-008, EXP-009) closed as *underpowered*, demonstrating bounded custody and systems behavior while withholding confirmatory promotion of structured retrieval effects on failure-prevention endpoints. We recommend explicit Mechanical Scientific Method discipline—custody receipts, claim ceilings, and terminal-state preservation—as baseline infrastructure for NewInML-style agent evaluations, without claiming that structured retrieval efficacy has been established.

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